

Joint Estimation Methodology

Fate

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1 Introduction

We are replacing the sequential alternating estimation with joint B_z , B_y estimation and will report the MAP and the covariance-variance matrix, Σ .

As usual, the process goes as

$$\begin{aligned} t = 0 \text{ of scan} &\rightarrow \text{Initialisation of prior} \rightarrow \text{Observation of data} \rightarrow \text{Bayes Update} \\ &\rightarrow \text{Kl Choice of } B_c \rightarrow \text{Repeat till } t = 70 \mu s \end{aligned}$$

2 Numerical Evaluvation of Bayes Update

1. **Initialisation of Prior** We have a 2D grid indexed by B_z , B_y containing the prior probability values.
2. **Evaluvation of Likelihood** We are yet to implement bias marginalisation. But otherwise the likelihood evaluvation is,

$$\ln \mathcal{L}(\mathbf{y}_{\text{exp}}^{(k)} | \mathbf{B}; B_c^{(k)}) = -\frac{1}{2\sigma^2} \left\| \mathbf{y}_{\text{exp}}^{(k)} - \mu_{\text{sim}}(\mathbf{B}; B_c^{(k)}) \right\|_2^2 + C,$$

where $\mathbf{B} = (B_z, B_y)$

3. **Bayes Update** Then we update the posterior by the Bayes Rule,

$$P(\mathbf{B} | \mathbf{y}_{1:k}, B_c^{(k)}) \propto \mathcal{L}(\mathbf{y}_{\text{exp}}^{(k)} | \mathbf{B}, B_c^{(k)}) \times P(\mathbf{B} | \mathbf{y}_{1:k-1}). \quad (1)$$

Now since we evaluvate the posterior weights on a 2D grid of fixed resolution, if the posterior width shrinks to the order of the grid spacing, we implement a zoom, to create a more fine resolved support for the posterior probability distribution, where we find out the region of $\sim 99\%$ probability mass, and consider that our new grid with greater resolution.

To assign probability values to the newly introduced grid points and to be rigorous, we use all the experimental data that the experiment has seen so far, and do a bayesian update starting with a uniform distribution to of the new grid.

We can explore interpolation of probability later if computation time becomes unbearable later.

4. **Bias Selection for the Next Step** We choose the longitudinal bias field for the next step, $B_c^{(k+1)}$ such that the expected KL Divergence Utility, $U(B_c)$ is maximised,

$$B_c^{(k+1)} = \arg \max_{B_c \in \mathcal{B}_c} U(B_c),$$

where,

$$U(B_c) = \int P(y | B_c) \times \left[\int P(\mathbf{B} | y, B_c) \ln \frac{P(\mathbf{B} | y, B_c)}{P(\mathbf{B} | \mathbf{y}_{1:k})} d\mathbf{B} \right] dy,$$

where,

$$P(y \mid B_c) = \int d\mathbf{B} P(y \mid B_c, \mathbf{B}) \cdot P(\mathbf{B} \mid \mathbf{y}_{1:k})$$

To keep this calculation numerically tractable, we do a monte carlo approximation of the integrals.

$$\begin{aligned} & \int dy d\mathbf{B}_1 P(y \mid B_c, \mathbf{B}_1) \cdot P(\mathbf{B}_1) \cdot \left[\int P(\mathbf{B} \mid y, B_c) \ln \frac{P(\mathbf{B} \mid y, B_c)}{P(\mathbf{B})} d\mathbf{B} \right] \\ & \approx \frac{\sum_{i \in [N]} }{N} \int P(\mathbf{B} \mid y_i, B_c) \ln \frac{P(\mathbf{B} \mid y_i, B_c)}{P(\mathbf{B})} d\mathbf{B} \end{aligned}$$

where, $\mathbf{B}_i$ are generated from $P(\mathbf{B})$ and y_i are generated from $P(y \mid B_c, \mathbf{B})$ and consequently, $(y_i, \mathbf{B}_i)$ are generated from the joint distribution.

Further, we can use the sample generated from $P(\mathbf{B})$ to approximate the other integral as well, as

$$\approx \frac{\sum_{i \in [N]} \sum_{j \in [N]} }{N^2} P(y_i \mid \mathbf{B}_j, B_c) \ln \frac{P(\mathbf{B}_j \mid y_i, B_c)}{P(\mathbf{B}_j)}.$$

5. And finally, repeat.

3 Results Seen So Far